

Contents

1	I. Introduction	1
1.1	Organization of The Report	2
2	II. Single-Diode Model	3
2.1	Simulating the single-diode model in LTspice	4
2.2	Solving the Single-Diode Model	5
3	III. The Likelihood Function	6
4	IV. Maximum Likelihood Estimation	7
4.1	Simulation	7
5	V. Confidence Interval with Bootstrapping	9
6	VI. Conclusion	11
	References	12

Chapter 1

I. Introduction

Performance is typically described using key parameters such as open-circuit voltage (VOC), short-circuit current (ISC), and maximum power output (Pmax). Although PV manufacturers can usually estimate these values with known uncertainty under standard reporting conditions (SRC), defined as a spectral irradiance of 1000 W/m^2 and a temperature of 25° C , evaluating performance outside of these conditions is more difficult. However, such assessments are crucial because photovoltaic devices commonly operate under a broad range of environmental conditions.

Characterization of such performance away from SRC may be done using mathematical models of the photovoltaic device. For calibration of such models, data from parameter estimation of I-V curve measurements are used. The estimation of such parameters can be increasingly difficult in some cases due to the small amount of data available, computational algorithms may have issues with the convergence, etc.

This report presents a single-diode model that describes the PV performance of a wide range of devices at irradiance values with the temperature fixed at 25° C . The estimates for the single diode model are obtained from the I-V curve data and it is desirable as it provides an estimate for the key performance parameters. The data for the I-V curve are simulated using circuit simulation software (LTspice) and the simulated data are used for the estimation of the key performance parameters. The formulation of the single diode model used in this report includes a statistical model which accounts for measurement of noise in irradiance and is equivalent to model parameter estimation using maximum likelihood estimation (MLE) techniques. The likelihood function is defined, incorporating a noise model for the points measured in I-V curve. An assumption is made that the designed model is estimable - i.e., a unique maximum of the likelihood function exists. Then the calculation of the maximum likelihood estimates of the single-diode model parameters are done for the simulated data set. After the parameters are estimated, we then calculate the 95% confidence interval for each of the estimated parameters using bootstrapping.

1.1 Organization of The Report

This chapter provides a background for the topics covered in this report. In Chapter 2, we provided a description of single-diode model and the procedure of modelling the diode for generating the key performance parameters. Then we use the data for estimating the parameters. For estimating these parameters we generate a likelihood function in Chapter 3. Gaussian noise is added in we also define the distribution of E in this chapter. The next chapter, Chapter 4, is based on the maximum likelihood estimation which is used for finding the estimates of the single-diode model parameters. In Chapter 5, we find the 95% confidence interval for the corresponding MLEs of the parameters. And finally, in chapter 6, we conclude with the maximum likelihood estimation of the parameters for a single diode model.

Chapter 2

II. Single-Diode Model

The single-diode lumped-parameter equivalent circuit model effectively captures the I-V behavior of many series-connected photovoltaic (PV) devices under typical irradiance levels and constant temperature conditions, for non-negative terminal voltages.[1]

$$I = I_L - I_S \left(e^{\frac{V+IR_S}{N_S n V_{th}}} - 1 \right) - \frac{V + IR_S}{R_P}, \quad (2.1)$$

where the thermal voltage is $V_{th} := k_B T/q$, and

I device terminal current (A)

V device terminal voltage (V)

I_L photocurrent (A)

I_S diode reverse saturation current of device (A)

N_S number of cells connected in series (unit-less)

n ideality factor of device (unit-less)

R_S series resistance of device (Ω)

R_P parallel (shunt) resistance of device (Ω)

k_B Boltzmann constant ($1.3806504 \times 10^{-23}$ J/K)

T junction temperature of device (K)

q electron charge ($1.602176487 \times 10^{-19}$ C).

In general, the five parameters— I_L , I_S , n , R_S , and R_P —may vary with temperature, irradiance, or terminal voltage. However, in this context, we assume no dependence on terminal voltage. Instead, we use auxiliary equations to describe how each parameter changes with temperature and irradiance relative to a reference value defined under standard reporting conditions (SRC). Since calibration lab measurements are typically conducted at a fixed temperature of $T = T_o = 25^\circ\text{C}$ and irradiance close to one sun, we assume that the effects of temperature and irradiance n , R_S , and R_P are minimal. Therefore, the auxiliary equations for these parameters are expressed accordingly.

$$\begin{aligned} n &= n_o \\ R_S &= R_{S_o} \\ R_P &= R_{P_o} \end{aligned}$$

where n_o diode ideality factor at SRC R_{S_o} series resistance at SRC R_{P_o} parallel resistance at SRC.

For $T = T_o = 25^\circ\text{C} = 298.15$ K, the photocurrent I_L is given by the auxiliary equation

$$I_L := EI_{SC_0} + I_0 \left(e^{\frac{EI_{SC_0} R_{S_0}}{N_S n_0 V_{th_0}}} - 1 \right) + \frac{EI_{SC_0} R_{S_0}}{R_{P_0}}. \quad (2.2)$$

The effective irradiance ratio E is a unit-less measure of the “effective” irradiance illuminating the device relative to 1 sun at SRC. The device’s short-circuit current at the prevailing irradiance, I_{SC} , divided by the device’s short-circuit current at SRC, I_{SC_0} , is known as E . The short-circuit current of a calibrated reference device with the same angular response and spectral matching to the device may be easily measured to determine this ratio. Random noise in E ’s measurement is a feature unique to the measurement testbed.

If the reference device is not spectrally matched, then from the definition of the unit-less spectral correction factor M [2], which compensates for spectral response differences between the device and the reference device, it follows that

$$E := \frac{I_{SC}}{I_{SC_0}} = \frac{M I_{SC,R}}{I_{SC,R_0}}. \quad (2.3)$$

where $I_{SC,R}$ is the reference device’s short-circuit current at the prevailing irradiance and I_{SC,R_0} is the reference device’s short-circuit current at SRC (typically a calibration value). This report does not consider the uncertainty due to M and I_{SC,R_0} . Auxiliary equation (2.2) describes the device’s photocurrent generation as a function of E and the model parameters. The equation is derived by setting $V = 0$ and $I = I_{SC}$ in (2.1), using definition (2.3) to substitute $I_{SC} = E I_{SC_0}$, and then solving for I_L . I_{L_0} can be derived directly from I_{SC,R_0} using (2.2) with $E = 1$. The model parameters which correspond to reference values at SRC are denoted by the vector $\theta = (I_{SC_0}, I_{S_0}, n_o, R_{S_0}, R_{P_0})$

2.1 Simulating the single-diode model in LTspice

Simulation of the single-diode model for I-V data used for estimation of different key performance parameters is done using LTspice. The schematic diagram for the single-diode model is given in 2.1.

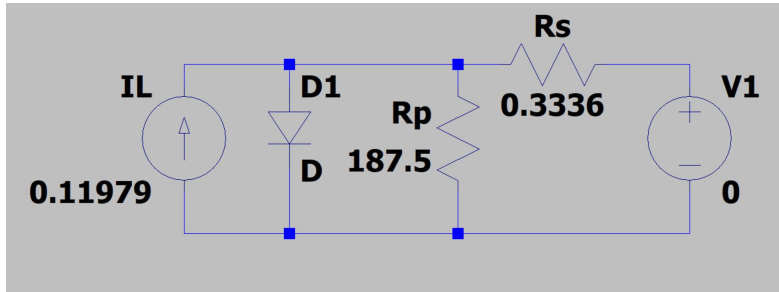


Fig. 2.1 Schematic for Single-Diode Model.

Simulation for the single-diode model is done by sweeping the DC voltage source from 0 to 1 V with an interval of 0.01. D1 is the 1N4148 diode used for the model. The values for I_S , N_S are set to 10^{-14} and 1.2, respectively, for the required simulation. Other values of the parameters are given as: $R_S = 0.3336$, $R_P = 187.5$ and $I_L = 0.11979$. By sweeping the DC voltage from 0-1V, we plot the $I(V1)$ vs $V1$ curve, which is further used for the maximum likelihood estimation of the key performance parameters.

2.2 Solving the Single-Diode Model

Solutions to the single-diode model (2.1) define a performance surface in (V, I, E) -space, given θ in parameter space $\Theta \subset \mathbb{R}^5$. This surface implicitly defines three functions: $I = g_I(V, E; \theta)$, $V = g_V(I, E; \theta)$ and $E = g_E(V, I; \theta)$. The computation of the I-V curve in the first quadrant is verified against explicit computation schemes using junction voltages in some papers over a large parameter space for θ and for a range of practical values for N_S . This ensures that the computational algorithms operate correctly for a wide variety of PV devices [3]. The I-V curve computed by using different samples of voltage and corresponding current is given in fig: 2.2.

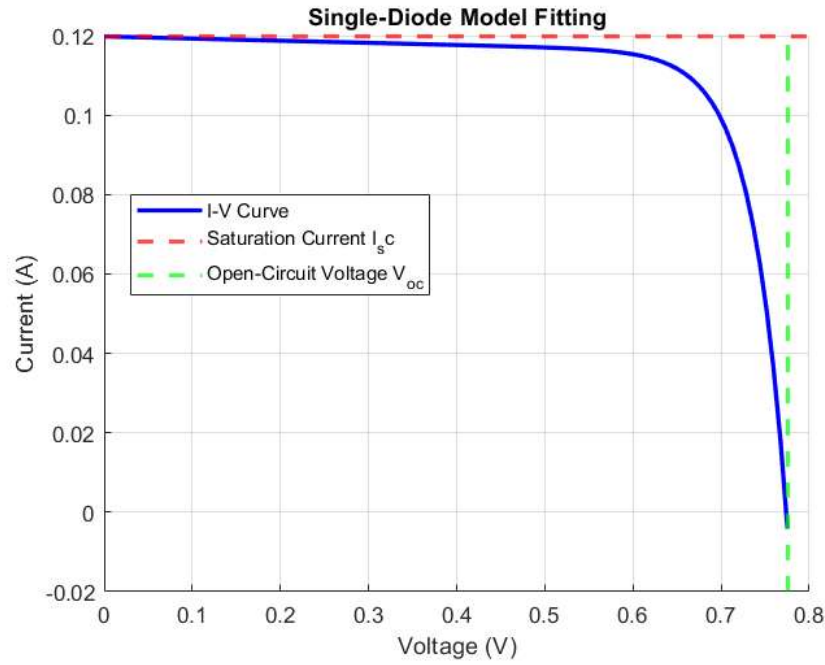


Fig. 2.2 I-V curve for the Single-Diode Model.

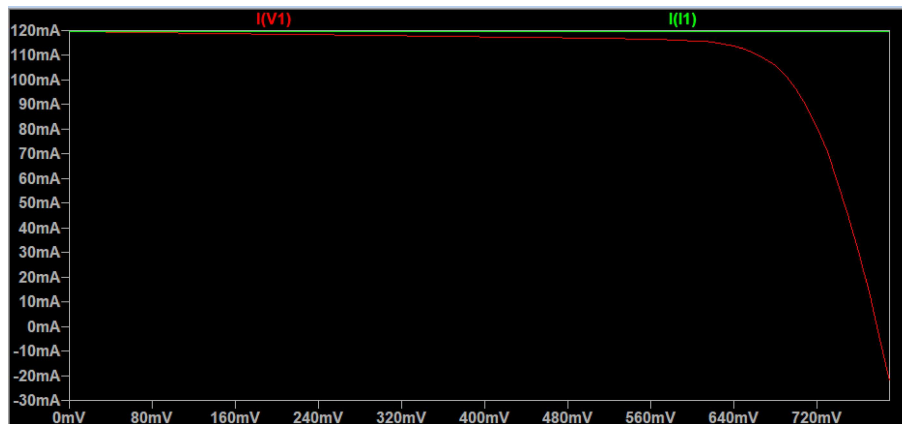


Fig. 2.3 I-V curve generated in LTspice.

Chapter 3

III. The Likelihood Function

We first explicitly state our assumptions while creating a probability function. First, we presume that the performance of the PV device at a given temperature close to one sun is well represented by the single-diode model (2.1)-(2.2). Secondly, a noise model for the observables has to be specified. In contrast to the measurement noise in E , we assume that the noise in the measurements of V and I is insignificant (this is a realistic assumption in many measurement circumstances). Therefore, given a value for, we take into consideration an observed effective irradiance value that is measured with noise but produced by the single-diode model (2.1) - (2.2) through the appropriate function $E = g_E(V, I; \boldsymbol{\theta})$. We assume that the noise in the measured E values follows a normal distribution. In other words, the normal random variable provides the statistical model for the value of the n th observable E_n for a sample of size N data points recorded along an I-V curve and for $n = 1, \dots, N$.

$$\mathcal{E}_n = g_E(V_n, I_n; \boldsymbol{\theta}) + \varepsilon_n,$$

where the additive noise ε_n is assumed to be independent and identically distributed (i.i.d) normally with zero mean and variance σ_E^2 , i.e., $\varepsilon_n \sim \mathcal{N}(0, \sigma_E^2)$. The likelihood function is defined from this assumption, and the corresponding MLE is obtained. When determining the probability, we see that, for $n = 1, \dots, N$, the statistical model provided determines the n_{th} measured effective irradiance value, which is a realization of the random variable, $\mathcal{E}_n$, which has a normal distribution.

$$\mathcal{E}_n \sim \mathcal{N}(g_E(V_n, I_n; \boldsymbol{\theta}), \sigma_E^2).$$

The likelihood function is derived by fixing the data to actual observed values, denoted $\mathcal{E} = \mathbf{E} = (E_1, \dots, E_N)$, and interpreting the joint probability density function (pdf) as a function of $\boldsymbol{\theta}$ and σ_E^2 . Since the data reassumed i.i.d., the likelihood function for the data is

$$L(\boldsymbol{\theta} \mid \mathcal{E}) = (2\pi\sigma_E^2)^{-N/2} \exp \left(-\frac{\sum_{n=1}^N (\mathcal{E}_n - g_E(V_n, I_n; \boldsymbol{\theta}))^2}{2\sigma_E^2} \right). \quad (3.1)$$

Chapter 4

IV. Maximum Likelihood Estimation

Given observed data $\mathbf{E}$, an MLE of the model parameters $\boldsymbol{\theta}$, denoted by $\hat{\boldsymbol{\theta}}_{ML}$, is value in the parameter space Θ that maximizes the likelihood function $L(\boldsymbol{\theta} \mid \mathcal{E})$:

$$\hat{\boldsymbol{\theta}}_{ML} = \arg \max_{\boldsymbol{\theta} \in \Theta} L(\boldsymbol{\theta} \mid \mathcal{E}). \quad (4.1)$$

We are also interested in finding an MLE for the key performance parameters at SRC: V_{OC0} , I_{SC0} and P_{max0} , along with finding an MLE for $\boldsymbol{\theta}$. We define $\boldsymbol{\theta}' := (I_{SC0}, I_{S0}, R_{P0}, R_{s0}, N_S, n, E, \sigma_E^2, V_{OC0})$ - which differs from $\boldsymbol{\theta}$ by the inclusion of E , σ_E^2 and V_{OC0} . Because we are assuming normal i.i.d noise in our data, we note that the MLE of $\boldsymbol{\theta}$ is equivalent to the nonlinear least squares estimate of $\boldsymbol{\theta}$. This is true because maximizing (4.1) with respect to $\boldsymbol{\theta}$ is equivalent to minimizing $\sum_{n=1}^N (\mathcal{E}_n - g_E(V_n, I_n; \boldsymbol{\theta}))^2$ with respect to $\boldsymbol{\theta}$. In addition, maximum likelihood estimation provides a method for quantifying our uncertainty in both $\boldsymbol{\theta}$ and $\boldsymbol{\theta}'$. An MLE is not guaranteed to exist nor is an MLE guaranteed to be unique if it does exist.

The basic procedure for finding MLEs of is the same as for finding MLEs of simpler problems: one uses an optimization technique to maximize the likelihood function. In simple cases such as estimating the mean and variance of data known to be from a normal distribution or in linear regression, one can find the MLE of the parameters by taking partial derivatives of the log likelihood function with respect to the parameters, setting those partial derivatives equal to zero, and solving the resulting system of equations over a particular domain. Since equation (4.1) is nonlinear and of higher dimension, we use non linear least square function `lscurvefit` to estimate and then use the corresponding functions of ML to estimate the key performance parameters.

4.1 Simulation

We define the single-diode model using the following user-defined function:

$$\begin{aligned} \text{single diode model} = & @(\theta, V) \dots \theta(1) - \theta(2) * (\exp((V + \theta(4) * \theta(1)) \\ & / (\theta(5) * \theta(6) * V_{th} * \theta(7))) - 1) - (V + \theta(4) * \theta(1)) / \theta(3); \end{aligned}$$

As we already defined $\boldsymbol{\theta}' := (I_{SC0}, I_{S0}, R_{P0}, R_{s0}, N_S, n, E, \sigma_E^2, V_{OC0})$, this vector is passed as an argument to the function `lscurvefit`, along with voltage (V) and $\mathcal{E}$ as other two arguments. The model is tested using the optimization method `optimoptions` with function and optimality tolerance of 10^{-8} each, maximum tolerance of 1000 and

step tolerance of 10^{-10} and $\sigma_E^2 = 10^{-6}$. We calculate the MLEs of the parameters for two different samples, one consisting of an initial guess as the true value (θ'_{true}) and the other sample consisting of the initial vector as the true value with some noise added to it. Table 4.1 shows the MLEs for two samples mentioned consisting of different parameters of the vector θ' .

Table 4.1 Maximum Likelihood Estimates of the Parameter

Parameter	Value θ'	Value θ' (with noise)
I_{SC0}	0.119790	0.119523
I_{S0}	2.2×10^{-10}	3.174×10^{-10}
R_{P0}	187.493833	208.0601
R_{S0}	0.3336	0.22662
N_S	1.00	0.85709
n	1.500002	1.5631
E	1.00	1.1183
σ_E^2	1×10^{-6}	1.110×10^{-6}
V_{oc}	0.775224	0.44602

The I-V curve for the estimated parameters and the simulated data are given in fig: 2.2

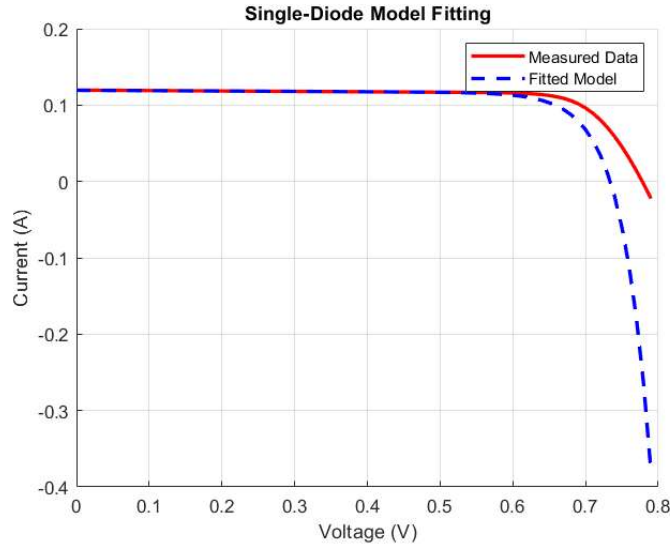


Fig. 4.1 I-V curve for the Single-Diode Model.

Chapter 5

V. Confidence Interval with Bootstrapping

Interval estimates quantify the uncertainty in point estimates by reporting a range of values that is likely to cover the true parameter θ' . Confidence intervals are typically used with the MLE. In this work we use the $(1-\alpha) \times 100\%$ confidence interval for the i^{th} parameter estimate, derived from the asymptotic distribution of $\hat{\theta}_{ML}$. Bootstrapping is a resampling technique that estimates the distribution of a parameter by repeatedly sampling with replacement from the observed data. This approach avoids making parametric assumptions about the distribution. For calculating the confidence interval using bootstrapping, we create `num_bootstrap = 1000` resampled datasets by sampling with replacement from $(V, \mathcal{E})$. The model is then fitted for each bootstrapped dataset using `lsqcurvefit`. To avoid storing bad results or crashing midway we use `try` and `catch ME` clause. The estimated parameters from each bootstrap iteration are stored in `bootstrap_estimates`. We then use the 95% confidence interval to obtain the confidence bounds using the 2.5^{th} and 97.5^{th} percentiles of the bootstrapped parameter estimates. For the given the 95% confidence interval is given in Table 5.1.

Table 5.1 Parameter Estimates with Confidence Intervals

Parameter	Value	95% Confidence Interval (CI)
I_{SC_0}	0.119523	(0.1195213, 0.1197228)
I_{S_0}	3.174×10^{-10}	(3.056×10^{-10} , 3.186×10^{-10})
R_{P_0}	208.0601	(187.4329, 210.4482)
R_{S_0}	0.22662	(0.22517, 0.228417)
N_S	0.85709	(0.857098, 0.859897)
n	1.5631	(1.56240, 1.56368)
E	1.1183	(1.11705, 1.11942)
σ_E^2	1.110×10^{-6}	(1.1100×10^{-6} , 1.11032×10^{-6})
V_{oc}	0.44602	(0.4460296, 0.4460306)

We add noise with E and calculate the value of $\mathcal{E}$ as $\mathcal{E}_n = g_E(V_n, I_n; \boldsymbol{\theta}) + \varepsilon_n$. The I-V plot for noisy data, $\mathcal{E}_n \sim \mathcal{N}(g_E(V_n, I_n; \boldsymbol{\theta}))$, and with the simulated values and for the parameters fitted in `lsqcurve` function is given in fig: 5.1

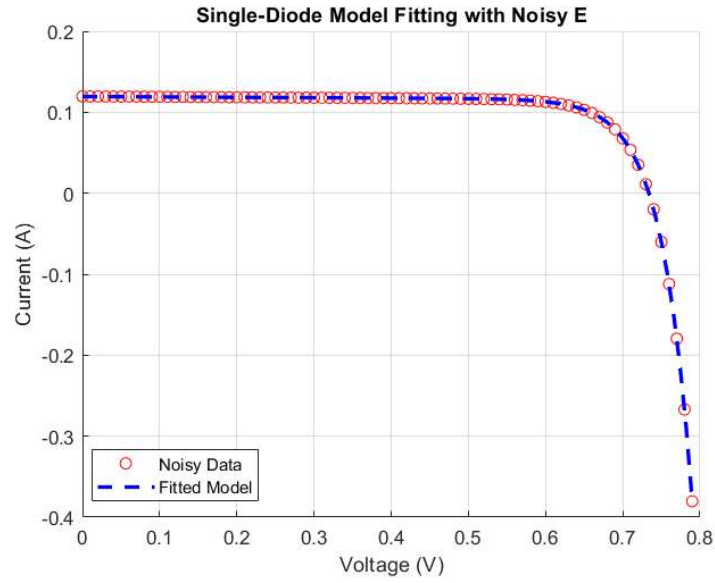


Fig. 5.1 I-V curve for the Single-Diode Model.

Chapter 6

VI. Conclusion

In this report we implemented a robust procedure to model the behavior of a photovoltaic (PV) cell using the single-diode model. We used LTspice for modelling the single-diode behaviour of a photovoltaic cell. Simulations were run to generate I-V curve for the model. Then by simulating theoretical I-V characteristics based on initial parameter guesses, including physical and environmental constants we plotted the I-V curve. We then used Maximum Likelihood Estimation for finding the estimates of the parameters using the I-V curve. Since for a gaussian distribution finding the MLE is equivalent to finding the non-linear least square estimates, experimental data was incorporated, and the nonlinear model was fitted using a least-squares optimization method (lsqcurvefit) to estimate key model parameters such as the photocurrent (I_L), saturation current (I_s), shunt and series resistances (R_p and R_s), ideality factor (n), and open-circuit voltage (V_{oc}).

To account for measurement noise and improve robustness, Gaussian noise was introduced in the model predictions, and the fitting was refined accordingly. The final model accurately replicates the measured I-V data, as evidenced by the close match between the fitted curve and the experimental points.

Furthermore, bootstrapping was used to assess the uncertainty of the estimated parameters. By generating multiple resampled datasets and re-fitting the model, the code computes 95% confidence intervals for each parameter. This adds significant statistical rigor to the analysis and enhances the reliability of the parameter estimates.

In summary, the report demonstrates a comprehensive and statistically sound methodology for PV cell parameter extraction, modeling, and validation, which can be valuable for both academic research and real-world solar panel diagnostics.

References

- [1] Brian Zaharatos, Mark Campanelli, Clifford Hansen, Keith Emery, and Luis Tenorio. Likelihood methods for single-diode model parameter estimation from noisy i–v curve data. In *2014 IEEE 40th Photovoltaic Specialist Conference (PVSC)*, pages 2850–2855. IEEE, 2014.
- [2] Carl R Osterwald. Translation of device performance measurements to reference conditions. *Solar cells*, 18(3):269–279, 1986.
- [3] JW Bishop. Computer simulation of the effects of electrical mismatches in photovoltaic cell interconnection circuits. *Solar cells*, 25(1):73–89, 1988.